STAT 4C03 Question 2

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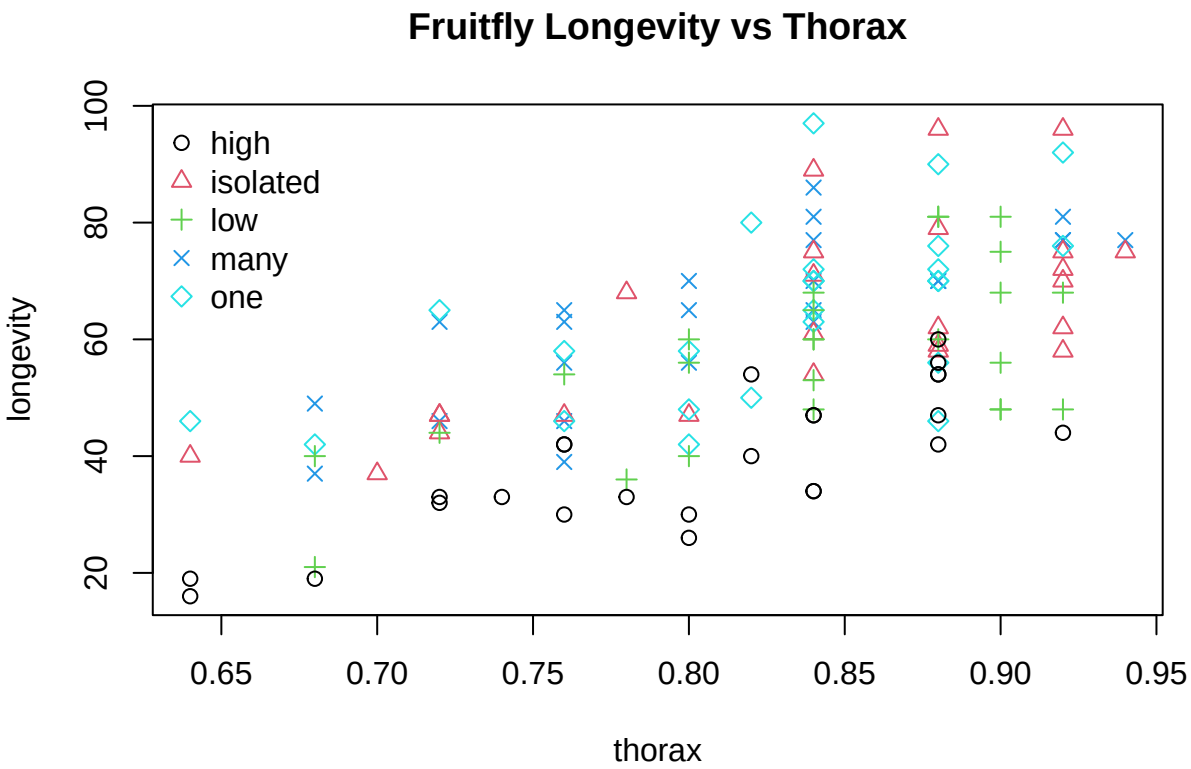
```
fruitfly = read.csv("Fruitfly.csv", stringsAsFactors = TRUE)
head(fruitfly)
```

```
##   thorax longevity activity
## 1   0.68        37     many
## 2   0.68        49     many
## 3   0.72        46     many
## 4   0.72        63     many
## 5   0.76        39     many
## 6   0.76        46     many
```

a)

```
plot(longevity ~ thorax, data = fruitfly,
     col  = as.integer(fruitfly$activity),
     pch  = as.integer(fruitfly$activity),
     main = "Fruitfly Longevity vs Thorax")

legend("topleft",
       legend = levels(fruitfly$activity),
       col    = 1:5,
       pch    = 1:5,
       bty    = "n")
```



There is a clear positive relationship between thorax and longevity in all five groups: larger individuals tend to live longer.

b)

```
m_int <- lm(longevity ~ thorax * activity, data = fruitfly)
anova(m_int)
```

```
## Analysis of Variance Table
##
## Response: longevity
##                  Df  Sum Sq Mean Sq F value    Pr(>F)
## thorax            1 15003.3 15003.3 130.733 < 2.2e-16 ***
## activity          4  9634.6  2408.6  20.988 5.503e-13 ***
## thorax:activity   4    24.3     6.1   0.053    0.9947
## Residuals       114 13083.0   114.8
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

conclusion:

We have a p-value = 0.99, there is no evidence whatsoever that the interaction is needed.

```
m_add <- lm(longevity ~ thorax + activity, data = fruitfly)
summary(m_add)
```

```
##
## Call:
## lm(formula = longevity ~ thorax + activity, data = fruitfly)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -26.108  -7.014  -1.101   6.234  30.265
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)       -68.753     10.401  -6.610 1.17e-09 ***
## thorax            134.341     12.731  10.552  < 2e-16 ***
## activityisolated   20.004      3.016   6.632 1.05e-09 ***
## activitylow        12.989      3.019   4.302 3.51e-05 ***
## activitymany       24.142      3.016   8.005 9.38e-13 ***
## activityone        22.641      2.999   7.550 1.01e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.54 on 118 degrees of freedom
## Multiple R-squared:  0.6527, Adjusted R-squared:  0.638
## F-statistic: 44.36 on 5 and 118 DF,  p-value: < 2.2e-16
```

```
anova(m_add, m_int)
```

```
## Analysis of Variance Table
##
## Model 1: longevity ~ thorax + activity
## Model 2: longevity ~ thorax * activity
##   Res.Df   RSS Df Sum of Sq     F Pr(>F)
## 1    118 13107
## 2    114 13083  4    24.314 0.053 0.9947
```

thorax: each additional mm of thorax length increases expected longevity by 134.3 days, holding activity group constant. The effect is highly significant and common to all groups.

activity (baseline = high:. 8 virgin females): all other groups live significantly longer than the high-virgin group:

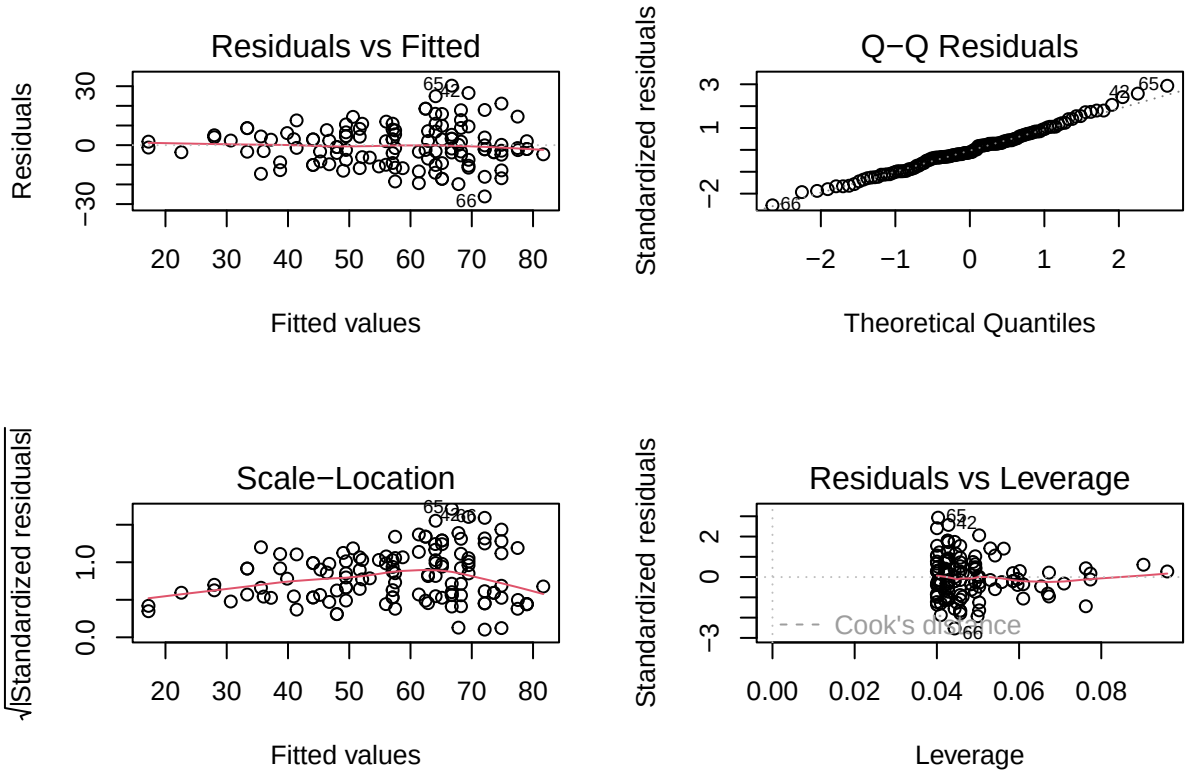
many (8 pregnant females): +24.1 days one (1 virgin female): +22.6 days isolated: +20.0 days low (1 pregnant female): +13.0 days

Overall fit:

R2=0.65R^2 = 0.65 R2=0.65, meaning the model explains 65% of the variation in longevity, which is reasonable for biological data.

c)

```
par(mfrow = c(2, 2))
plot(m_add)
```



Residuals vs Fitted: the spread of residuals increases as fitted values get larger, this is a fan shape, indicating that flies with longer predicted lifespans also have more variable lifespans.

Normal Q-Q: the points follow the line reasonably well in the middle but deviate at the tails, suggesting the residuals are slightly right-skewed. Normality is not perfectly met.

Main issues: non-constant variance and mild non-normality.

d)

```
fruitfly$log_long <- log(fruitfly$longevity)

m_log <- lm(log_long ~ thorax + activity, data = fruitfly)
summary(m_log)
```

```
##
## Call:
## lm(formula = log_long ~ thorax + activity, data = fruitfly)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.52641 -0.13629 -0.00823  0.13918  0.39273
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)        1.42496    0.19059   7.477 1.47e-11 ***
## thorax             2.72146    0.23329  11.666  < 2e-16 ***
## activityisolated   0.41925    0.05527   7.586 8.35e-12 ***
```

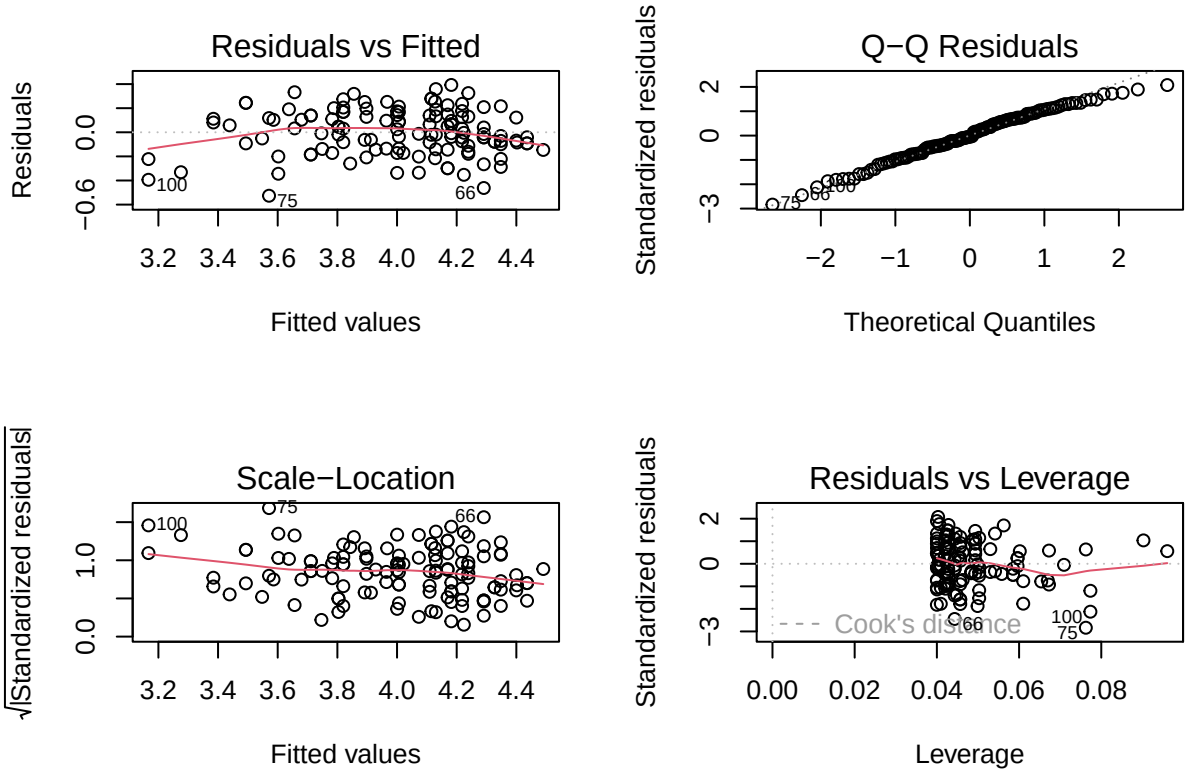
```
## activitylow       0.29538    0.05532   5.339 4.59e-07 ***
## activitymany      0.50716    0.05527   9.176 1.78e-15 ***
## activityone       0.47100    0.05495   8.571 4.62e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1931 on 118 degrees of freedom
## Multiple R-squared:  0.7025, Adjusted R-squared:  0.6899
## F-statistic: 55.72 on 5 and 118 DF,  p-value: < 2.2e-16
```

thorax: exp(2.72) ~= 15.2, each additional mm of thorax multiplies expected longevity by about 15, holding activity constant. The effect is proportional rather than additive. activity (baseline = high): all coefficients are interpreted as multiplicative factors.

R2 improves from 0.653 to 0.703, the log model explains more variance.

The log transformation fixes heteroscedasticity and improves normality.

```
par(mfrow = c(2, 2))
plot(m_log)
```



e)

```
m_gamma <- glm(longevity ~ thorax + activity, family = Gamma(link = "log"), data = fruitfly)
summary(m_gamma)
```

```
##
## Call:
## glm(formula = longevity ~ thorax + activity, family = Gamma(link = "log"),
##     data = fruitfly)
##
```

```
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)        1.47256    0.18601   7.916 1.49e-12 ***
## thorax             2.68778    0.22769  11.804  < 2e-16 ***
## activityisolated   0.41466    0.05394   7.687 4.93e-12 ***
## activitylow        0.29820    0.05400   5.523 2.02e-07 ***
## activitymany       0.49716    0.05394   9.217 1.43e-15 ***
## activityone        0.46993    0.05363   8.762 1.66e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Gamma family taken to be 0.0355297)
##
##     Null deviance: 13.2803  on 123  degrees of freedom
## Residual deviance:  4.3151  on 118  degrees of freedom
## AIC: 942.29
##
## Number of Fisher Scoring iterations: 4
```

```
# Side-by-side coefficient comparison
cbind(
  lm_log   = coef(m_log),
  gamma    = coef(m_gamma)
)
```

```
##                     lm_log     gamma
## (Intercept)      1.4249578 1.4725639
## thorax           2.7214581 2.6877811
## activityisolated 0.4192521 0.4146590
## activitylow      0.2953848 0.2981985
## activitymany     0.5071617 0.4971552
## activityone      0.4709964 0.4699328
```

```
cbind(
  se_lm  = sqrt(diag(vcov(m_log))),
  se_glm = sqrt(diag(vcov(m_gamma)))
)
```

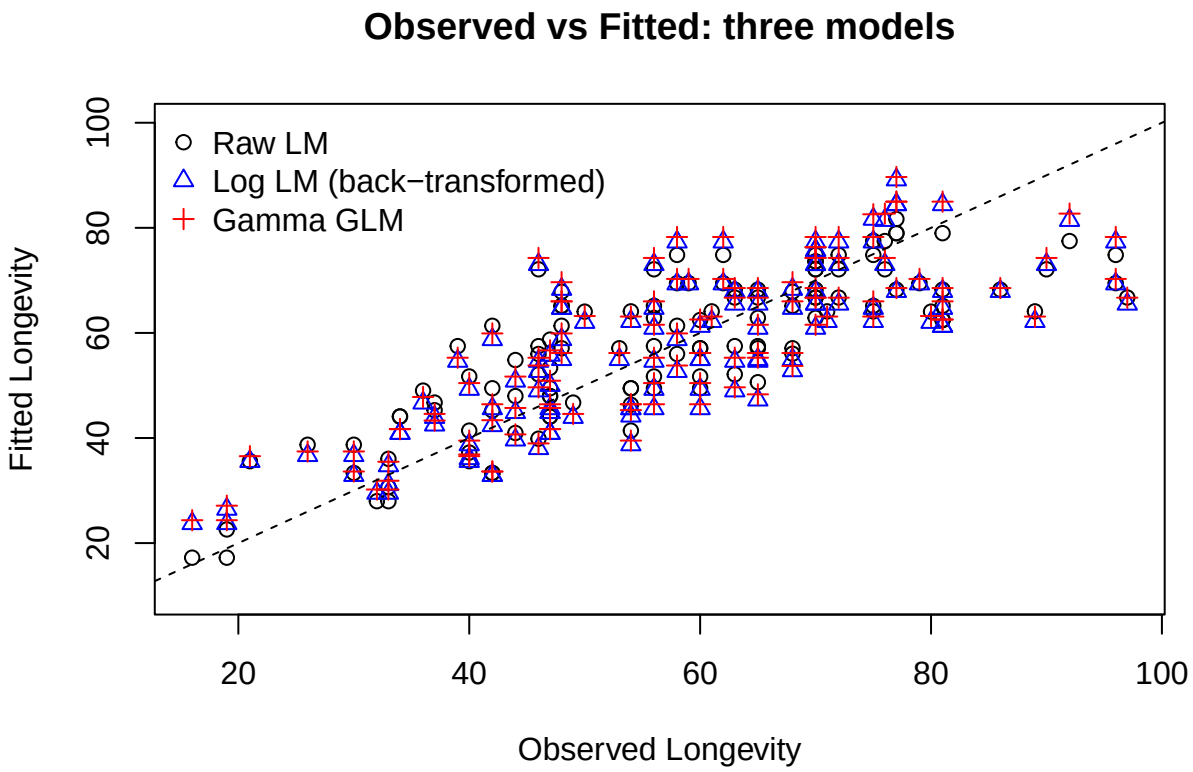
```
##                      se_lm     se_glm
## (Intercept)      0.19058651 0.18601403
## thorax           0.23328932 0.22769233
## activityisolated 0.05526631 0.05394039
## activitylow      0.05532427 0.05399695
## activitymany     0.05526752 0.05394157
## activityone      0.05494997 0.05363163
```

The coefficient estimatesand stanard errors are very similar between the two models.

f)

```r
fitted_raw   <- fitted(m_add)
fitted_log   <- exp(fitted(m_log))      # back-transform to original scale
fitted_gamma <- fitted(m_gamma)

plot(fruitfly$longevity, fitted_raw,   pch = 1, col = "black",
     xlab = "Observed Longevity", ylab = "Fitted Longevity",
     main = "Observed vs Fitted: three models", ylim = c(10, 100))
points(fruitfly$longevity, fitted_log,   pch = 2, col = "blue")
points(fruitfly$longevity, fitted_gamma, pch = 3, col = "red")
abline(0, 1, lty = 2)
legend("topleft", legend = c("Raw LM", "Log LM (back-transformed)", "Gamma GLM"),
       col = c("black", "blue", "red"), pch = 1:3, bty = "n")
```



The **Gamma GLM** (red crosses) and **log LM** (blue triangles) produce nearly identical fitted values across the entire range, the two symbols overlap almost perfectly throughout the plot. Both track the diagonal well, showing no systematic bias.

The **raw LM** (black circles) deviates more noticeably: it tends to **overestimate** for low observed values and **underestimate** for high ones, reflecting the heteroscedasticity issue identified in part (c).

Overall, the Gamma GLM and log LM are essentially interchangeable in terms of fitted values, with the raw LM being the weakest of the three.